India's TECH ECOSYSTEM Encompassing IoT Deployment Is At CURIOUS CROSSROADS

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UNISOC is a fabless semiconductor company committed to R&D of core chipsets in mobile communications and AIoT. Its products cover mobile chipset platforms supporting 2G/3G/4G/5G communication standards and various chipset solutions in the field of IoT, RFFE, wireless connection, TV, etc. Eric Zhou, executive vice president, UNISOC, throws light on the ways India based IoT firms can scale-up deployment and results

Q. Which are the top three technologies that are shaping the future of IoT? How are these influencing the future of IoT?

A. The growth trajectory of the global IoT ecosystem is being shaped by several interconnected, interdependent technologies.

Take, for instance, artificial intelligence (AI) and related technologies such as machine learning and data analytics. Not only are these technologies providing the framework on which IoT solutions are built but are also helping in ensuring better service delivery, higher personalisation, and more interconnectivity to the benefit of the end-consumer.

The mass deployment of many advanced IoT solutions is hindered by bandwidth-related challenges. The introduction of 5G will address these issues and allow for high-speed connectivity needed to power the IoT ecosystem, leading to the development and deployment of the next wave of platform architectures and devices.

The biggest driver for IoT, however, will be NB-IoT. This technology, which leverages DSS modulation, is being hailed as a faster and potentially less expensive IoT connectivity option despite its higher upfront costs. This is because, unlike other wireless connectivity options, NB-IoT allows sensor data to be directly communicated to the primary server.

Q. Between the cellular and NB-IOT/LPWA families of wireless technologies, which one are you betting upon? Why?

A. We have a broad range of good products in NB-IoT/CAT1/GPRS. All of these directions have good conditions in industrial ecology, and we will be focussing on these. UNISOC is one of the few companies in the industry that can provide the most comprehensive low-power wide-area (LPWA) technology portfolio.

Q. How ready is India's tech eco-system to develop and deploy IoT solutions?

A. When it comes to IoT deployment, India's tech ecosystem is at curious-crossroads. On the one hand, we have telecommunication companies that exist in the IoT workflow across almost all solution development stages because of their advantages in network ownership. On the other hand, however, telecommunication players in the nascent IoT ecosystem still have to deal with several critical challenges such as standardisation, interoperability, and data security.

This is further complicated by the fact that low-power, low-cost channels such as NB technology are yet to find traction in India. So, while the ecosystem shows promise, there are many measures that need to be taken to ensure that the proper rules, processes, procedures, audits, accountability, and coherence are in place.

Q. Do you foresee India's tech industry developing its own IP and branded products/solutions in the IoT arena?

A. There is definitely a huge scope for Indian companies to develop their own IPs and branded solutions/products in the IoT space. This is because of the scope of disruption that is available to them—primarily in the IoT products space, which currently is not as versatile as the consumer segment.

Partnering with leading chipset manufacturers to localise these value propositions for the Indian market will help indigenous companies generate unique IPs and brand differentiation.

Q. Do you see the open source phenomenon playing an important role in the IoT arena?

A. Any technology moves through four cycles of market acceptance, and open source development plays a vital role in helping it gain market traction. Why? Because open source applications help in exploring new use-cases for technologies to make them more viable for end-consumers, business processes, and industries. The IoT space is no exception to this paradigm.

That said, it will be vital to introduce certain regulations that monitor the adoption of open-source IoT products and solutions in large-scale applications. IoT manufacturers also need to explore stronger linkages across the supply chain—especially with systems integrators, connectivity solutions providers, and chipset manufacturers—with a security-first perspective. These measures will help in addressing the various security vulnerabilities and challenges that currently restrict the domain.

Q. What are the key technologies missing, which when made available will accelerate the adoption of IoT across the globe?

A. 5G's big connection is now happening. 5G's low-latency application may be a key technology in IoT. We predict that ultra-reliable low latency communication (URLLC) will be truly deployed in many places within three years; at the same time, various types of sensors are also IoT outbreaks. EFY